



Lesson 1-1

An Introduction to Complex Numbers

مقدمة في الأعداد المركبة

ملزمة الطالب | Student Handout



امسح الباركود للدخول للمنصة
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1. Pre-requirements -- متطلبات سابقة

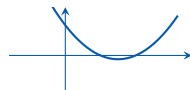
قبل الدخول في العدد المركب، نراجع سريعاً حل المعادلة التربيعية -- لأن العدد المركب ه يظهر بالظبط في الحالة التي معندهاش حل حقيقي.

First: solving $ax^2 + bx + c = 0$ ($a \neq 0$) algebraically — the general formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

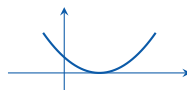
The quantity $\Delta = b^2 - 4ac$ is called the **discriminant**; its sign tells us the type of roots *before* we even finish solving.

Second: solving $ax^2 + bx + c = 0$ graphically. Let $f(x) = ax^2 + bx + c$ and sketch the curve $y = f(x)$. The real roots of the equation are exactly the x -coordinates of the points where the curve meets the x -axis. According to the sign of Δ , three cases arise:



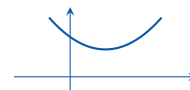
$$\Delta > 0$$

two distinct real roots



$$\Delta = 0$$

one repeated real root



$$\Delta < 0$$

curve never meets the x -axis — **no real root**

When $\Delta < 0$ the equation still has *two* solutions — they are just not real numbers. وهنا بالظبط هبدأ درسنا النهاردة: today we build the number system, $\mathbb{C}$, that contains them.

2. Review of Key Definitions -- استعراض التعريفات

- **Imaginary unit:** since $x^2 + 1 = 0$ has no real solution, we define i so that

$$i^2 = -1 \quad (\text{equivalently } i = \sqrt{-1})$$

- **Powers of i repeat every 4 steps:** $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, then it repeats.
- **Complex number:** $z = a + bi$, where $a, b \in \mathbb{R}$. $a = \text{Re}(z)$ is the *real part*, $b = \text{Im}(z)$ is the *imaginary part*.

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i^2 = -1\}$$

- **Equality of two complex numbers:** $a + bi = c + di \iff a = c \text{ and } b = d$.
- **Conjugate:** of $z = a + bi$ is $\bar{z} = a - bi$. Two useful facts: $z + \bar{z} = 2a \in \mathbb{R}$ and $z \cdot \bar{z} = a^2 + b^2 \in \mathbb{R}^+$.

3. Lesson Explanation -- شرح الدرس

- Some quadratics (e.g. $x^2 + 4 = 0$) have a **negative discriminant** $\Rightarrow$ no real roots. We solve them by writing $\sqrt{-k} = i\sqrt{k}$ for $k > 0$ — يعني: نطلع الإشارة السالبة بـ i الجذر على هيئة i الأول، وبعدين نبسط الجذر العادي.
- **Quick rule for i^n :** divide n by 4 and look at the remainder r :

$$r = 0 \Rightarrow 1, \quad r = 1 \Rightarrow i, \quad r = 2 \Rightarrow -1, \quad r = 3 \Rightarrow -i$$

- **Arithmetic:** add/subtract real parts together and imaginary parts together; multiply by expanding normally and replacing i^2 by -1 ; divide by multiplying the numerator and denominator by the *conjugate of the denominator* (this always turns the denominator into a real number) — الهدف من ضرب البسط والمقام في مرافق المقام إننا نتخلص من i في المقام خالص.

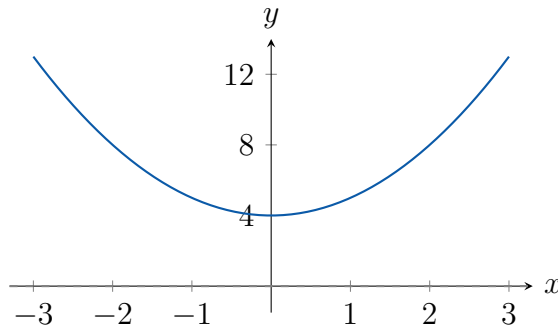
Exam-saving identity: $(1 + i)^2 = 2i$ and $(1 - i)^2 = -2i$.

So $(1 + i)^{2n} = (2i)^n$ and $(1 - i)^{2n} = (-2i)^n$ — this turns a big power into a one-line calculation.

4. Worked Examples -- الأمثلة

Example 1 — Graph -- رسم بياني

Sketch $y = x^2 + 4$ for $x \in [-3, 3]$. Use the graph to explain why $x^2 + 4 = 0$ has no real solution, then solve the equation in $\mathbb{C}$.



Solution / الحل

- 1 Lowest point of the curve: $(0, 4)$ أقل نقطة على المنحنى فوق محور x
- 2 The curve never meets the x -axis $\implies$ no real solution الحل الحقيقي = نقطة تقاطع مع محور x ، ومفيش تقاطع
- 3 $x^2 + 4 = 0 \implies x^2 = -4$ نحل في $\mathbb{C}$
- 4 $x = \pm\sqrt{-4} = \pm 2i$
- 5 Solution set = $\{2i, -2i\}$

Answer: $\{2i, -2i\}$

Example 2 — Essay

Write $\sqrt{-45}$ in the simplest form.

Solution / الحل

- 1 $\sqrt{-45} = \sqrt{45} \cdot \sqrt{-1}$ نطلع الإشارة السالبة برة الجذر
- 2 $= \sqrt{45} \cdot i$ لأن $\sqrt{-1} = i$
- 3 $= \sqrt{9 \cdot 5} \cdot i$ نبسط الجذر العادي
- 4 $= 3\sqrt{5}i$

Answer: $3\sqrt{5}i$

Example 3 — Essay -- تنبيه لأشهر غلطةFind the value of $\sqrt{-8} \times \sqrt{-2}$.**Solution / الحل**

- 1 Common mistake: $\sqrt{-8} \times \sqrt{-2} = \sqrt{16} = 4$ (wrong) أشهر غلطة: القاعدة $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$ للأعداد غير السالبة بس
- 2 $\sqrt{-8} = \sqrt{8}i = 2\sqrt{2}i$ نحول كل جذر لصورة i الأول
- 3 $\sqrt{-2} = \sqrt{2}i$
- 4 $\sqrt{-8} \times \sqrt{-2} = (2\sqrt{2}i)(\sqrt{2}i)$
- 5 $= 2(2)i^2 = 4i^2$ $\sqrt{2} \cdot \sqrt{2} = 2$
- 6 $= 4(-1) = -4$

Answer: -4 **Example 4 — Essay**Find the simplest form of $i^{75} + i^{-58}$.**Solution / الحل**

- 1 $75 = 4(18) + 3 \implies i^{75} = i^3 = -i$ نقسم الأس على 4 وناخذ الباقي
- 2 $58 = 4(14) + 2 \implies i^{58} = i^2 = -1$
- 3 $i^{-58} = \frac{1}{i^{58}} = \frac{1}{-1} = -1$ الأس السالب معناه المقلوب
- 4 $i^{75} + i^{-58} = -i + (-1) = -1 - i$

Answer: $-1 - i$

Example 5 — Essay -- صيغة رمزية

If n is a positive integer, find the simplest form of i^{4n+1} .

Solution / الحل

$$1 \quad i^{4n+1} = i^{4n} \cdot i^1$$

نفصل الأس

$$2 \quad = (i^4)^n \cdot i$$

$$3 \quad = (1)^n \cdot i$$

لأن $i^4 = 1$

$$4 \quad = i$$

لأي عدد صحيح موجب n

Answer: i

Example 6 — MCQ

$$(1 - i)^6 = \dots\dots\dots$$

$$(a) \ 8i$$

$$(b) \ -8i$$

$$(c) \ 8$$

$$(d) \ -8$$

Solution / الحل

$$1 \quad (1 - i)^2 = -2i$$

المتطابقة التي بتوفّر وقت الامتحان

$$2 \quad (1 - i)^6 = [(1 - i)^2]^3 = (-2i)^3$$

$$3 \quad = -8i^3$$

$$4 \quad = -8(-i) = 8i$$

لأن $i^3 = -i$

Answer: (a) $8i$

Example 7 — Essay

Find the real numbers x, y satisfying: $(2x - 1) + (3y + 2)i = 5 - 4i$.

Solution / الحل

$$1 \quad 2x - 1 = 5$$

نساوي الحقيقي بالحقيقي

$$2 \quad 2x = 6 \implies x = 3$$

$$3 \quad 3y + 2 = -4$$

والتخيلي بالتخيلي

$$4 \quad 3y = -6 \implies y = -2$$

$$5 \quad (x, y) = (3, -2)$$

Answer: $(x, y) = (3, -2)$

Example 8 — Essay -- قيمة مجهول

Find the real value of m so that $z = (m + 4) + (2m - 6)i$ is a **purely imaginary** number.

Solution / الحل

$$1 \quad \text{Purely imaginary} \implies \text{Re}(z) = 0 \text{ and } \text{Im}(z) \neq 0$$

$$2 \quad m + 4 = 0 \implies m = -4$$

الجزء الحقيقي = صفر

$$3 \quad \text{Check: } \text{Im}(z) = 2(-4) - 6 = -14 \neq 0$$

لازم نتأكد إن الجزء التخيلي مش صفر

$$4 \quad m = -4$$

Answer: $m = -4$

Example 9 — EssaySolve in $\mathbb{C}$: $x^2 - 4x + 13 = 0$.**Solution / الحل**

1 $a = 1, \quad b = -4, \quad c = 13$

2 $\Delta = (-4)^2 - 4(1)(13) = 16 - 52 = -36$

المميز سالب، يبقى الجذران مركبان

3 $x = \frac{4 \pm \sqrt{-36}}{2}$

القانون العام

4 $= \frac{4 \pm 6i}{2}$

لأن $\sqrt{-36} = 6i$

5 $= 2 \pm 3i$

6 Solution set = $\{2 + 3i, 2 - 3i\}$

Answer: $\{2 + 3i, 2 - 3i\}$ **Example 10 — Essay**If $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, find in the form $a + bi$: (a) $z_1 + z_2$ (b) $z_1 \cdot z_2$.**Solution / الحل**

1 (a) $z_1 + z_2 = (3 - 1) + (-2 + 4)i$

حقيقي مع حقيقي، ونحلي مع تخلي

2 $= 2 + 2i$

3 (b) $z_1 \cdot z_2 = (3 - 2i)(-1 + 4i)$

4 $= -3 + 12i + 2i - 8i^2$

نفك الأقواس

5 $= -3 + 14i + 8$

نعوض $i^2 = -1$

6 $= 5 + 14i$

Answer: (a) $2 + 2i$ (b) $5 + 14i$

Example 11 — MCQ

If $z = 4 - 3i$, then $z + \bar{z} = \dots\dots\dots$

- (a) 8 (b) $-6i$
(c) $8 - 6i$ (d) 25

Solution / الحل

1 $\bar{z} = 4 + 3i$

المرافق: نغيّر إشارة الجزء التخيلي بس

2 $z + \bar{z} = (4 - 3i) + (4 + 3i)$

3 $= 8$

Answer: (a) 8

Example 12 — Essay -- اكتشف الخطأ

Sara and Omar were asked to simplify $\sqrt{-9} \times \sqrt{-16}$.

Sara: $\sqrt{-9} \times \sqrt{-16} = \sqrt{(-9)(-16)} = \sqrt{144} = 12$.

Omar: $\sqrt{-9} \times \sqrt{-16} = (3i)(4i) = 12i^2 = -12$.

Which student is correct? Explain Sara's mistake.

Solution / الحل

1 Omar is correct.

2 $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$ only when $a \geq 0$ and $b \geq 0$ سارة استخدمت القاعدة دي — وهي للأعداد غير السالبة بس

3 $\sqrt{-9} = 3i$, $\sqrt{-16} = 4i$ العددان سالبان، فنحوّل كل جذر لصورة i الأول

4 $\sqrt{-9} \times \sqrt{-16} = (3i)(4i) = 12i^2$

5 $= 12(-1) = -12$

Answer: Omar: -12

Example 13 — Essay -- برهان

If $a, b \in \mathbb{R}$ and $a + bi = \frac{3 + i}{1 - 2i}$, prove that $a^2 + b^2 = 2$.

Proof / البرهان

$$1 \quad \frac{3 + i}{1 - 2i} \times \frac{1 + 2i}{1 + 2i}$$

نضرب البسط والمقام في مرافق المقام

$$2 \quad = \frac{3 + 6i + i + 2i^2}{1 + 4}$$

$$3 \quad = \frac{1 + 7i}{5}$$

نعوض $i^2 = -1$

$$4 \quad = \frac{1}{5} + \frac{7}{5}i \implies a = \frac{1}{5}, \quad b = \frac{7}{5}$$

$$5 \quad a^2 + b^2 = \frac{1}{25} + \frac{49}{25}$$

$$6 \quad = \frac{50}{25} = 2 \quad \blacksquare$$

Answer: $a^2 + b^2 = 2$

Example 14 — Essay

Put $\frac{5 - i}{2 + 3i}$ in the form $a + bi$.

Solution / الحل

$$1 \quad \frac{5 - i}{2 + 3i} \times \frac{2 - 3i}{2 - 3i}$$

نضرب البسط والمقام في مرافق المقام

$$2 \quad = \frac{10 - 15i - 2i + 3i^2}{4 + 9}$$

المقام: عدد $\times$ مرافقه $= a^2 + b^2$

$$3 \quad = \frac{10 - 17i - 3}{13}$$

نعوض $i^2 = -1$

$$4 \quad = \frac{7 - 17i}{13}$$

$$5 \quad = \frac{7}{13} - \frac{17}{13}i$$

نفصل الحقيقي عن التخيلي

Answer: $\frac{7}{13} - \frac{17}{13}i$

Example 15 — MCQWhich of the following equals a **real** number?

(a) $(2 + i)(2 - i)$

(b) $(3 + 2i) - (3 - 2i)$

(c) i^5

(d) $\sqrt{-9}$

Solution / الحل

1 (a) $(2 + i)(2 - i) = 4 - i^2 = 4 + 1 = 5$

نحسب كل اختيار لوحده – حقيقي

2 (b) $(3 + 2i) - (3 - 2i) = 4i$

مش حقيقي

3 (c) $i^5 = i$

مش حقيقي

4 (d) $\sqrt{-9} = 3i$

مش حقيقي

Answer: (a) $(2 + i)(2 - i)$ **Example 16 — Essay -- تحدد**Find the simplest form of $(1 + i)^{10}$.**Solution / الحل**

1 $(1 + i)^2 = 2i$

المتطابقة الي بتوفر وقت الامتحان

2 $(1 + i)^{10} = [(1 + i)^2]^5 = (2i)^5$

3 $= 32i^5$

4 $= 32i$

لأن $i^5 = i$ **Answer:** $32i$ **Example 17 — Essay -- تطبيق**

Two resistances are connected in a circuit. The current in the first is $(6 - 2i)$ amperes and in the second is $(-1 + 5i)$ amperes. If the total current is the sum of the two currents, find the total current in the form $a + bi$.

Solution / الحل

1 Total = $(6 - 2i) + (-1 + 5i)$

التيار الكلي = مجموع التيارين

2 $= (6 - 1) + (-2 + 5)i$

3 $= 5 + 3i$ amperes

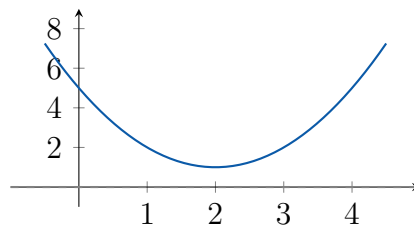
Answer: $5 + 3i$ amperes

5. Additional Exercises -- تمارين إضافية

حاول تحل كل تمرين بنفسك الأول. الحلول المفصلة خطوة بخطوة لكل التمارين دي موجودة في آخر الملزمة، في قسم Detailed Solutions.

Part A — Multiple choice

1. $i^{34} = \dots$ (a) i (b) -1 (c) $-i$ (d) 1
2. All are real except i^{17} (a) $(3+2i)+(3-2i)$ (b) i^{40} (c) $(2+i)(2-i)$ (d) $(4+i)i$
3. If $(x+1) + 3i = 5 + (y-2)i$, then $(x, y) = \dots$ (a) $(4, 5)$ (b) $(6, 1)$ (c) $(4, 1)$ (d) $(6, 5)$
4. $(2i)^3 = \dots$ (a) $8i$ (b) $-8i$ (c) 8 (d) -8
5. The conjugate of $-3 + 5i$ is $3-5i$ (a) $-3-5i$ (b) $3+5i$ (c) $5i-3$ (d) $5i+3$
6. If $z = 1 + i$, then $z\bar{z} = \dots$ (a) 0 (b) 1 (c) 2 (d) $2i$
7. Solution set of $x^2 + 9 = 0$ in $\mathbb{C}$ is $\{3i\}$ (a) $\{-3, 3\}$ (b) $\{3i\}$ (c) $\{3i, -3i\}$ (d) $\{3i, -3i\}$
8. Which product is real? $(4+i)(1+i)$ (a) $(4+i)(4-i)$ (b) $(4+i)^2$ (c) $(4+i)i$ (d) $(4-i)i$
9. $i^{-1} = \dots$ (a) i (b) $-i$ (c) 1 (d) -1
10. $(1+i)^4 = \dots$ (a) 4 (b) -4 (c) $4i$ (d) $-4i$
11. If $a + bi = 0$ ($a, b \in \mathbb{R}$), then $a = 0$ only (a) $b = 0$ only (b) $a = b = 0$ (c) $a = -b$ (d) $a = -b$
12. $i^{2026} = \dots$ (a) 1 (b) -1 (c) i (d) $-i$
13. $\sqrt{-64} = \dots$ (a) $8i$ (b) $-8i$ (c) $\pm 8i$ (d) 8
14. $(\sqrt{-3})(\sqrt{-27}) = \dots$ (a) 9 (b) -9 (c) $9i$ (d) $-9i$
15. If n is a positive integer, $i^{4n+3} = \dots$ (a) 1 (b) -1 (c) i (d) $-i$
16. From the graph of $y = x^2 - 4x + 5$ below, the equation $x^2 - 4x + 5 = 0$ has $\dots$ real root(s).



- (a) 0 (b) 1 (c) 2 (d) infinitely many

Part B — Essay questions

(1) Simplify to the simplest form:

(i) $i^{45} + i^{50}$

(ii) $(2 - i)^2$

(iii) $(1 + i)^8$

(iv) $3i^2 + 5i^3 - 2i^4$

(2) Put in the form $a + bi$:

$$(i) \frac{7 + 4i}{2i}$$

$$(ii) \frac{3 + 2i}{1 - i}$$

$$(iii) \frac{1 - 3i}{1 + 3i}$$

(3) Solve in $\mathbb{C}$:

$$(i) x^2 + 25 = 0$$

$$(ii) x^2 - 6x + 10 = 0$$

$$(iii) 3x^2 + 27 = 0$$

$$(iv) x^2 + 2x + 5 = 0$$

(4) Find the real numbers x, y satisfying:

$$(i) (3x - 2) + (y + 4)i = 7 - i$$

$$(ii) (x + y) + (x - y)i = 6 + 2i$$

$$(iii) (2 + i)(x + yi) = 4 + 7i$$

(5) Find the real value of k so that $z = (2k - 6) + (k + 1)i$ is a **purely imaginary number.**

(6) Mona simplified $(\sqrt{-4})(\sqrt{-9})$ as $\sqrt{36} = 6$. Explain the mistake in her reasoning and find the correct value.

6. Quiz on Lesson 1–1 -- كويز على الدرس

(Total: 10 marks)

First question

6 marks (1 mark each)

Choose the correct answer:

1. $i^{58} = \dots\dots$ (a) 1 (b) -1 (c) i (d) $-i$
2. If $z = 2 - 3i$, then $\bar{z} = \dots\dots$ (a) $2+3i$ (b) $-2+3i$ (c) $-2-3i$ (d) $3-2i$
3. Solution set of $x^2 + 16 = 0$ in $\mathbb{C}$ is $\dots\dots$ (a) $\{4\}$ (b) $\{-4, 4\}$ (c) $\{4i\}$ (d) $\{4i, -4i\}$
4. $(1 - i)^2 = \dots\dots$ (a) $2i$ (b) $-2i$ (c) 2 (d) -2
5. Which is a *purely imaginary* number? $\dots\dots$ (a) 5 (b) $-7i$ (c) $3+4i$ (d) $5-2i$
6. If $x, y \in \mathbb{R}$ and $(x - 1) + (2y + 3)i = 0$, then $(x, y) = \dots\dots$ (a) $(1, -\frac{3}{2})$ (b) $(-1, \frac{3}{2})$ (c) $(1, \frac{3}{2})$ (d) $(0, 0)$

Second question

4 marks

[a] 2 marks

[b] 2 marks

[a] Solve in $\mathbb{C}$: $x^2 - 2x + 5 = 0$.

Solution / الحل:

[b] If $a, b \in \mathbb{R}$ and $a + bi = \frac{4 + 3i}{1 - i}$, find the value of $a + b$.

Solution / الحل:

7. Homework -- الواجب

To be solved and submitted -- يُحل ويُسلَّم

1. (MCQ) $i^{101} = \dots\dots\dots$ (a) 1 (b) -1 (c) i (d) $-i$
2. (MCQ) The conjugate of the real number 6 is (a) 6 (b) -6 (c) $6i$ (d) $-6i$
3. (Essay) Simplify: $(3 + 2i) - (5 - 4i) + (1 + i)$
4. (Essay) Simplify: $(2 + 3i)(1 - 2i)$
5. (MCQ) Which equation has *complex-conjugate* roots? (a) $x^2 - 4 = 0$ (b) $x^2 - 4x + 4 = 0$ (c) $x^2 + 4 = 0$ (d) $x^2 - 4x = 0$
6. (Essay) Solve in $\mathbb{C}$: $x^2 + 6x + 13 = 0$
7. (Essay) Find the real x, y : $(x + 2) + (y - 1)i = 3 - 5i$
8. (Essay) Put in the form $a + bi$: $\frac{6 - 2i}{3 + i}$
9. (Essay) If $z = 3 + 4i$, show that $z\bar{z}$ is a positive real number and find its value.
10. (MCQ) The graph of $y = x^2 + 2x + 5$ (a) touches the x -axis once (b) crosses it twice (c) never meets the x -axis (d) passes through the origin

8. Detailed Solutions -- الحلول المفصلة للتمارين الإضافية

راجع حلك هنا بعد ما تحاول. كل خطوة في سطر، وجنبا توضيح بالعربي عند اللزوم.

Part A — Multiple choice

No. 1

$$i^{34} = \dots\dots$$

- (a) i (b) -1 (c) $-i$ (d) 1

Solution / الحل

$$1 \quad 34 = 4(8) + 2$$

نقسم الأس على 4 وناخذ الباقي

$$2 \quad i^{34} = i^2 = -1$$

Answer: (b) -1

No. 2

All are real except

- (a) $(3 + 2i) + (3 - 2i)$ (b) i^{40} (c) $(2 + i)(2 - i)$ (d) i^{17}

Solution / الحل

$$1 \quad (a) \quad (3 + 2i) + (3 - 2i) = 6$$

نحسب كل اختيار لوحده – حقيقي

$$2 \quad (b) \quad 40 = 4(10) + 0 \implies i^{40} = 1$$

حقيقي

$$3 \quad (c) \quad (2 + i)(2 - i) = 4 + 1 = 5$$

عدد $\times$ مرافقه – حقيقي

$$4 \quad (d) \quad 17 = 4(4) + 1 \implies i^{17} = i$$

مش حقيقي

Answer: (d) i^{17}

No. 3

If $(x + 1) + 3i = 5 + (y - 2)i$, then $(x, y) = \dots\dots$

- (a) $(4, 5)$ (b) $(6, 1)$ (c) $(4, 1)$ (d) $(6, 5)$

Solution / الحل

$$1 \quad x + 1 = 5 \implies x = 4$$

الحقيقي بالحقيقي

$$2 \quad 3 = y - 2 \implies y = 5$$

والتخيلي بالتخيلي

$$3 \quad (x, y) = (4, 5)$$

Answer: (a) $(4, 5)$

No. 4

$(2i)^3 = \dots\dots$

- (a)
- $8i$
- (b)
- $-8i$
- (c)
- 8
- (d)
- -8

Solution / الحل

$$1 \quad (2i)^3 = 2^3 \cdot i^3 = 8i^3$$

$$2 \quad = 8(-i) = -8i$$

$$\text{لأن } i^3 = -i$$

Answer: (b) $-8i$ **No. 5**The conjugate of $-3 + 5i$ is

- (a)
- $3 - 5i$
- (b)
- $-3 - 5i$
- (c)
- $3 + 5i$
- (d)
- $5i - 3$

Solution / الحل

$$1 \quad z = -3 + 5i \implies \bar{z} = -3 - 5i$$

نغير إشارة الجزء التخيلي بس

Answer: (b) $-3 - 5i$ **No. 6**If $z = 1 + i$, then $z\bar{z} = \dots\dots$

- (a)
- 0
- (b)
- 1
- (c)
- 2
- (d)
- $2i$

Solution / الحل

$$1 \quad z\bar{z} = a^2 + b^2$$

عدد $\times$ مرافقه

$$2 \quad = 1 + 1 = 2$$

Answer: (c) 2 **No. 7**Solution set of $x^2 + 9 = 0$ in $\mathbb{C}$ is

- (a)
- $\{3\}$
- (b)
- $\{-3, 3\}$
- (c)
- $\{3i\}$
- (d)
- $\{3i, -3i\}$

Solution / الحل

$$1 \quad x^2 + 9 = 0 \implies x^2 = -9$$

$$2 \quad x = \pm\sqrt{-9} = \pm 3i$$

$$3 \quad \text{Solution set} = \{3i, -3i\}$$

Answer: (d) $\{3i, -3i\}$

No. 8

Which product is real?

- (a)
- $(4 + i)(1 + i)$
- (b)
- $(4 + i)(4 - i)$
- (c)
- $(4 + i)^2$
- (d)
- $(4 + i)i$

Solution / الحل

1 (b) $(4 + i)(4 - i) = 16 + 1 = 17$

عدد × مرافقه = عدد حقيقي دائماً

Answer: (b) $(4 + i)(4 - i)$ **No. 9** $i^{-1} = \dots\dots$

- (a)
- i
- (b)
- $-i$
- (c)
- 1
- (d)
- -1

Solution / الحل

1 $i^{-1} = \frac{1}{i}$

الأس السالب معناه المقلوب

2 $= \frac{1}{i} \times \frac{i}{i} = \frac{i}{i^2}$

نضرب البسط والمقام في i

3 $= \frac{i}{-1} = -i$

Answer: (b) $-i$ **No. 10** $(1 + i)^4 = \dots\dots$

- (a)
- 4
- (b)
- -4
- (c)
- $4i$
- (d)
- $-4i$

Solution / الحل

1 $(1 + i)^4 = [(1 + i)^2]^2 = (2i)^2$

لأن $(1 + i)^2 = 2i$

2 $= 4i^2 = -4$

Answer: (b) -4 **No. 11**If $a + bi = 0$ ($a, b \in \mathbb{R}$), then

- (a)
- $a = 0$
- only (b)
- $b = 0$
- only (c)
- $a = b = 0$
- (d)
- $a = -b$

Solution / الحل

1 $a + bi = 0 + 0i$

الصفر عدد مركب جزؤه الحقيقي صفر وجزؤه التخيلي صفر

2 $a = 0$ and $b = 0$

الحقيقي بالحقيقي، والتخيلي بالتخيلي

Answer: (c) $a = b = 0$

No. 12

$i^{2026} = \dots\dots$

- (a) 1 (b)
- -1
- (c)
- i
- (d)
- $-i$

Solution / الحل

1 $2026 = 4(506) + 2$

نقسم الأس على 4 ونأخذ الباقي

2 $i^{2026} = i^2 = -1$

Answer: (b) -1

No. 13

$\sqrt{-64} = \dots\dots$

- (a)
- $8i$
- (b)
- $-8i$
- (c)
- $\pm 8i$
- (d) 8

Solution / الحل

1 $\sqrt{-64} = \sqrt{64} \cdot \sqrt{-1}$

2 $= 8i$

Answer: (a) $8i$

No. 14

$(\sqrt{-3})(\sqrt{-27}) = \dots\dots$

- (a) 9 (b)
- -9
- (c)
- $9i$
- (d)
- $-9i$

Solution / الحل

1 $\sqrt{-3} = \sqrt{3}i, \quad \sqrt{-27} = 3\sqrt{3}i$

نحول كل جذر لصورة i الأول

2 $(\sqrt{3}i)(3\sqrt{3}i) = 9i^2$

$\sqrt{3} \cdot 3\sqrt{3} = 9$

3 $= -9$

Answer: (b) -9

No. 15

If n is a positive integer, $i^{4n+3} = \dots\dots$

- (a) 1 (b)
- -1
- (c)
- i
- (d)
- $-i$

Solution / الحل

1 $i^{4n+3} = i^{4n} \cdot i^3$

نفصل الأس

2 $= (1)^n \cdot (-i)$

لأن $i^3 = -i$ و $i^4 = 1$

3 $= -i$

Answer: (d) $-i$

No. 16

From the graph of $y = x^2 - 4x + 5$, the equation $x^2 - 4x + 5 = 0$ has real root(s).

- (a) 0 (b) 1 (c) 2 (d) infinitely many

Solution / الحل

- | | |
|--|-----------------------|
| 1 Lowest point of the curve (vertex): (2, 1) | أقل نقطة فوق محور x |
| 2 The curve never reaches the x -axis | مفیش أي نقطة تقاطع |
| 3 Number of real roots = 0 | |

Answer: (a) 0

Part B — Essay questions

(1) (i)

Simplify to the simplest form: $i^{45} + i^{50}$

Solution / الحل

- | | |
|---|-------------------------------|
| 1 $45 = 4(11) + 1 \implies i^{45} = i$ | نقسم كل أس على 4 ونأخذ الباقي |
| 2 $50 = 4(12) + 2 \implies i^{50} = i^2 = -1$ | |
| 3 $i^{45} + i^{50} = i - 1 = -1 + i$ | |

Answer: $-1 + i$

(1) (ii)

Simplify to the simplest form: $(2 - i)^2$

Solution / الحل

- | | |
|------------------------------|--------------------|
| 1 $(2 - i)^2 = 4 - 4i + i^2$ | مربع مقدار ذي حدین |
| 2 $= 4 - 4i - 1$ | نعوض $i^2 = -1$ |
| 3 $= 3 - 4i$ | |

Answer: $3 - 4i$

(1) (iii)

Simplify to the simplest form: $(1 + i)^8$

Solution / الحل

$$1 \quad (1 + i)^8 = [(1 + i)^2]^4 = (2i)^4 \quad \text{لأن } (1 + i)^2 = 2i$$

$$2 \quad = 16i^4$$

$$3 \quad = 16 \quad \text{لأن } i^4 = 1$$

Answer: 16

(1) (iv)

Simplify to the simplest form: $3i^2 + 5i^3 - 2i^4$

Solution / الحل

$$1 \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$

$$2 \quad 3i^2 + 5i^3 - 2i^4 = 3(-1) + 5(-i) - 2(1)$$

$$3 \quad = -3 - 5i - 2$$

$$4 \quad = -5 - 5i$$

Answer: $-5 - 5i$

(2) (i)

Put in the form $a + bi$: $\frac{7 + 4i}{2i}$

Solution / الحل

$$1 \quad \frac{7 + 4i}{2i} \times \frac{-i}{-i} \quad \text{مرافق } 2i \text{ هو } -2i, \text{ فيكفي نضرب في } -i$$

$$2 \quad = \frac{-7i - 4i^2}{2} \quad \text{المقام: } 2i \times (-i) = -2i^2 = 2$$

$$3 \quad = \frac{4 - 7i}{2} \quad \text{نعوض } i^2 = -1$$

$$4 \quad = 2 - \frac{7}{2}i$$

Answer: $2 - \frac{7}{2}i$

(2) (ii)

Put in the form $a + bi$: $\frac{3 + 2i}{1 - i}$

Solution / الحل

$$1 \quad \frac{3 + 2i}{1 - i} \times \frac{1 + i}{1 + i}$$

نضرب البسط والمقام في مرافق المقام

$$2 \quad = \frac{3 + 3i + 2i + 2i^2}{2}$$

المقام: $1 + 1 = 2$

$$3 \quad = \frac{1 + 5i}{2}$$

نعوض $i^2 = -1$

$$4 \quad = \frac{1}{2} + \frac{5}{2}i$$

Answer: $\frac{1}{2} + \frac{5}{2}i$

(2) (iii)

Put in the form $a + bi$: $\frac{1 - 3i}{1 + 3i}$

Solution / الحل

$$1 \quad \frac{1 - 3i}{1 + 3i} \times \frac{1 - 3i}{1 - 3i}$$

نضرب البسط والمقام في مرافق المقام

$$2 \quad = \frac{1 - 6i + 9i^2}{10}$$

المقام: $1 + 9 = 10$

$$3 \quad = \frac{-8 - 6i}{10}$$

نعوض $i^2 = -1$

$$4 \quad = -\frac{4}{5} - \frac{3}{5}i$$

Answer: $-\frac{4}{5} - \frac{3}{5}i$

(3) (i)

Solve in $\mathbb{C}$: $x^2 + 25 = 0$

Solution / الحل

1 $x^2 = -25$

2 $x = \pm\sqrt{-25} = \pm 5i$

3 Solution set = $\{5i, -5i\}$

Answer: $\{5i, -5i\}$

(3) (ii)

Solve in $\mathbb{C}$: $x^2 - 6x + 10 = 0$

Solution / الحل

1 $\Delta = (-6)^2 - 4(1)(10) = 36 - 40 = -4$

المميز سالب

2 $x = \frac{6 \pm \sqrt{-4}}{2} = \frac{6 \pm 2i}{2}$

القانون العام

3 $= 3 \pm i$

4 Solution set = $\{3 + i, 3 - i\}$

Answer: $\{3 + i, 3 - i\}$

(3) (iii)

Solve in $\mathbb{C}$: $3x^2 + 27 = 0$

Solution / الحل

1 $3x^2 = -27 \implies x^2 = -9$

نقسم الطرفين على 3

2 $x = \pm\sqrt{-9} = \pm 3i$

3 Solution set = $\{3i, -3i\}$

Answer: $\{3i, -3i\}$

(3) (iv)

Solve in $\mathbb{C}$: $x^2 + 2x + 5 = 0$

Solution / الحل

$$1 \quad \Delta = (2)^2 - 4(1)(5) = 4 - 20 = -16$$

المميز سالب

$$2 \quad x = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2}$$

القانون العام

$$3 \quad = -1 \pm 2i$$

$$4 \quad \text{Solution set} = \{-1 + 2i, -1 - 2i\}$$

Answer: $\{-1 + 2i, -1 - 2i\}$

(4) (i)

Find the real numbers x, y satisfying: $(3x - 2) + (y + 4)i = 7 - i$

Solution / الحل

$$1 \quad 3x - 2 = 7 \implies 3x = 9 \implies x = 3$$

الحقيقي بالحقيقي

$$2 \quad y + 4 = -1 \implies y = -5$$

والتخيلي بالتخيلي

$$3 \quad (x, y) = (3, -5)$$

Answer: $(x, y) = (3, -5)$

(4) (ii)

Find the real numbers x, y satisfying: $(x + y) + (x - y)i = 6 + 2i$

Solution / الحل

$$1 \quad x + y = 6, \quad x - y = 2$$

نساوي الحقيقي بالحقيقي والتخيلي بالتخيلي

$$2 \quad 2x = 8 \implies x = 4$$

نجمع المعادلتين

$$3 \quad 4 + y = 6 \implies y = 2$$

$$4 \quad (x, y) = (4, 2)$$

Answer: $(x, y) = (4, 2)$

(4) (iii)

Find the real numbers x, y satisfying: $(2 + i)(x + yi) = 4 + 7i$

Solution / الحل

$$1 \quad (2 + i)(x + yi) = 2x + 2yi + xi + yi^2 \quad \text{نك الأقراس الأول}$$

$$2 \quad = (2x - y) + (x + 2y)i \quad \text{نعوض } i^2 = -1 \text{ ونجمع}$$

$$3 \quad 2x - y = 4, \quad x + 2y = 7 \quad \text{نساوي بالطرف الأيمن}$$

$$4 \quad y = 2x - 4 \implies x + 2(2x - 4) = 7 \implies 5x = 15 \implies x = 3 \quad \text{نحل المعادلتين}$$

$$5 \quad y = 2(3) - 4 = 2$$

$$6 \quad (x, y) = (3, 2)$$

Answer: $(x, y) = (3, 2)$

(5)

Find the real value of k so that $z = (2k - 6) + (k + 1)i$ is a **purely imaginary** number.

Solution / الحل

$$1 \quad \text{Purely imaginary} \implies \text{real part} = 0$$

$$2 \quad 2k - 6 = 0 \implies k = 3$$

$$3 \quad \text{Check: } k + 1 = 4 \neq 0 \quad \text{الجزء التخيلي مش صفر، فالقيمة مقبولة}$$

$$4 \quad k = 3$$

Answer: $k = 3$

(6)

Mona simplified $(\sqrt{-4})(\sqrt{-9})$ as $\sqrt{36} = 6$. Explain the mistake in her reasoning and find the correct value.

Solution / الحل

$$1 \quad \sqrt{a} \cdot \sqrt{b} = \sqrt{ab} \quad \text{only when } a \geq 0 \text{ and } b \geq 0 \quad \text{منى استخدمت القاعدة دي - وهي للأعداد غير السالبة بس}$$

$$2 \quad \sqrt{-4} = 2i, \quad \sqrt{-9} = 3i \quad \text{العددان سالبان، فنحوّل كل جذر لصورة } i \text{ الأول}$$

$$3 \quad (\sqrt{-4})(\sqrt{-9}) = (2i)(3i) = 6i^2$$

$$4 \quad = 6(-1) = -6$$

Answer: -6